

Projectiles

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This package includes a versatile and consistent projectile system to make various kinds of attacks. It is applicable to most types of attacks, However, The scripts that currently exist are not applicable to making hitscan attacks (you can probably get close with a very high projectile speed, but it could have issues if the projectile speed is too high), orbiting projectiles and homing projectiles.

ProjectileInfo

The core information of the projectile, contains information on the player who owns the projectile, team the projectile belongs to, the layers it should interact with, and object it should spawn when its SpawnObject method is called.

ProjectileGroup

Projectiles can be grouped by putting them inside a parent object with ProjectileGroup script. Projectile group is very useful for creating attacks that fire multiple projectiles at once, such as shotgun or cone wave style of attack. Note that the parent object of a group of projectiles needs to have ProjectileInfo as well as ProjectileGroup script

ProjectilePhysics

Handles physics operations of the projectile, causing it to move in local forward Z direction and falling in global Y downward direction. Every fixed update, the projectile first does a sphere cast in the direction of its movement to check what the projectile hit between the current position and the position after the fixed update. The information on what the projectile hit is then sent to the ProjectileInfo script for other projectile interaction scripts to use.

This script does not support projectile changing direction outside of the effect of gravity. If you need to make a projectile that orbits a point or homes to a location, you will need to create a similar script with custom projectile movement and physics scanning.

AreaOfEffectCircle

Handles an effect applied in an area, intended to be used for explosion or other types of area of effect abilities. It functions similarly to ProjectilePhysics script otherwise.

DestroyOnCollision

Causes the associated projectile to be destroyed when colliding with a certain object. What it should be destroyed from can be set for teammate, enemies, and environment.

ApplyEffect

Abstract class for applying an effect when colliding with a certain object. Like DestroyOnCollision, the object it interacts with can be set to self, teammate, and enemies. It uses the information on what object the projectile hit from the ProjectileInfo, then uses it to effect whoever the projectile hits.

See ApplyDamage, ApplyDisplacement, ApplyHealing, and ApplyStatus for examples of child class of ApplyEffect.

Note that projectile group parents also has projectileInfo, meaning the ApplyEffect can be used for the group to make a group of projectiles apply an effect generally and avoid having multiple hits of projectile apply effects independently